Wet antenna attenuation in commercial microwave links: a dynamic law identified from data

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Commercial microwave links can measure rain, but a film of water on the antenna covers adds an attenuation that is not caused by rain on the path. This wet antenna attenuation (WAA) is largest, relative to the signal, on short links, and it is often removed with a constant correction or by subtracting an assumed rain law. We identify a dynamic WAA law from the OpenMRG data of Gothenburg without subtracting any rain law. The antenna part and the path part are separated by their different dependence on the link length, and the WAA alone is measured in the minutes after rain when the radar sees no rain on the path. The resulting equation is a fast and a slow water store driven by the rain rate, $dW_f/dt = (0.72\,u - W_f)/\tau_f$ with $\tau_f = 2.6$ min while wetting and 10.3 min while drying, a slow store with 240 min, a target $u = a_{\mathrm{link}} A_{\mathrm{band}} (R/5\ \mathrm{mm/h})^{0.54}$, and a path loss $L\,c_{\mathrm{band}} R$. On 18 held-out storms, read once after a hashed freeze, the equation predicts the attenuation better than no WAA, a constant WAA, a WAA without memory, the two published WAA models with their default parameters and the refitted published model without memory, and as well as the refitted published model with memory. Of all models with a link factor it is the only one whose link factor does not change with the link length. Its main limit is the WAA level in light rain, which on the held-out storms is 0.4 to 0.6 dB higher than the equation predicts.

1. INTRODUCTION

Mobile networks connect their towers with point-to-point radio links, called commercial microwave links (CMLs). Each link is a beam between two dish antennas, typically a few kilometres apart, at tens of GHz. The radios record the transmitted and the received signal level, TSL and RSL in dBm, and their difference is the attenuation along the path. Raindrops absorb and scatter microwaves, so the attenuation rises when it rains on the path, and the dense existing networks can serve as rain sensors [1]. The measured attenuation, after the dry reference level (the baseline) is removed, is written as

$$A(t) = \mathrm{TSL}(t) - \mathrm{RSL}(t) - B = A_{\mathrm{rain}}(t) + \delta(t), \quad (1)$$

where B is the baseline and $\delta(t)$ collects every term that is not rain on the path. The rain term is a power law integrated along the path,

$$A_{\mathrm{rain}}(t) = \int_0^L a\,R(x,t)^b\,dx \approx a\,\bar{R}(t)^b\,L, \quad (2)$$

with R the rain rate in mm/h, $\bar{R}$ its mean along the path, L the link length in km, and a and b coefficients that depend on frequency and polarisation [2]. The approximation is exact only for uniform rain.

Rain also wets the covers of the two antennas (the radomes), and a water film on a radome attenuates the signal. This wet antenna attenuation, written W in dB, is the main part of $\delta(t)$ during rain. It is local to the antennas, so it does not grow with L, and it has memory, because the film builds up after rain starts and dries after rain stops. On a short link the path term is small, so the same W is a large share of A. A rain estimate that ignores W overestimates the rain, most strongly on short links [3], and an antenna that is still wet after the rain makes the link report rain that has already ended. Figure 1 shows one storm on a 0.36 km and on a 4.13 km link.

Existing WAA models are of two kinds. Schleiss *et al.* [4] describe a WAA that rises toward a maximum while it rains and decreases exponentially after the rain, measured on one experimental link. Pastorek *et al.* [5] compared practical corrections, among them a WAA that depends on the rain rate directly and has no memory. Direct measurements of the radome in the ATTRRA2 experiment showed that the real dynamics can be more complex than these models [6], and laboratory measurements showed water running off a radome directly after wetting [7].

A tempting way to measure WAA is to subtract a power-law rain term, Eq. (2), from A and to call the rest WAA. We do not do this. Every error of the assumed rain law would then be counted as WAA, and a constant WAA would look like a path term $f = L + L_0$, so the later study of the path law (Part 2 of the project) would inherit the error. The aim of this work is a readable, dynamic equation for $W(t)$ with the rain as input, found from real data in a way that never assumes a rain law. The work contributes four things. First, a model-free length test that measures the antenna and the path parts separately from the dependence on L. Second, a two-store WAA equation with wetting and drying times measured with rain gauges, whose antenna level and path loss were learned as free curves before closed forms were chosen. Third, three tests, which need no rain law, that the WAA does not depend on the link length. Fourth, one comparison of all WAA models under one protocol on held-out storms, both in accuracy and in physical consistency.

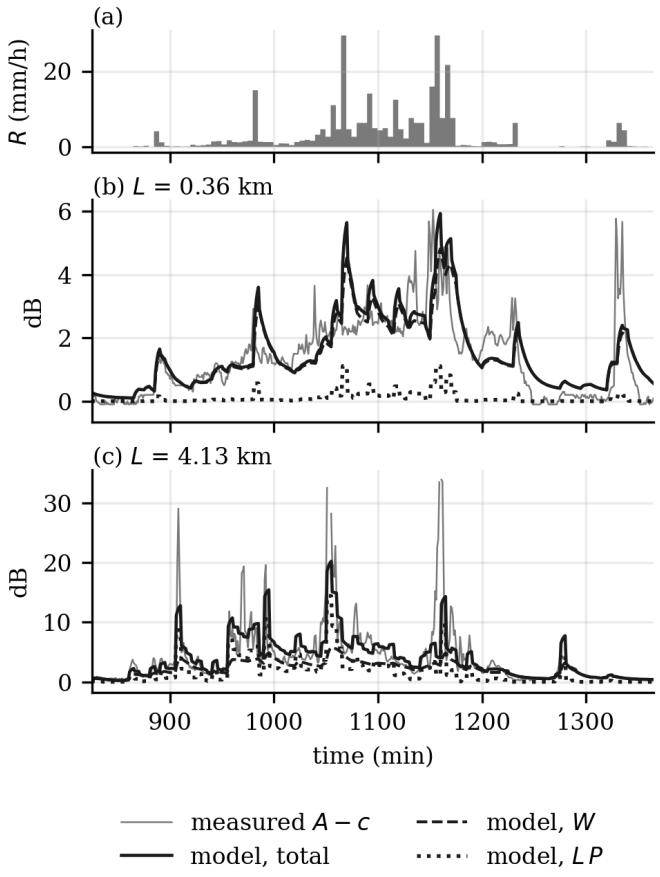


FIG. 1. One training storm on a short and a long 28 GHz link, one value per minute. (a) Rain over the short link. (b), (c) Measured attenuation $A - c$ (thin grey line) and the model of Sec. 4.2: total (solid), its antenna part W (dashed) and its path part $L\,P$ (dotted). On the short link the antenna part is most of the signal.

2. DATA AND PREPROCESSING

2.1. OpenMRG

We use OpenMRG, an open dataset from Gothenburg, Sweden, for 1 June to 31 August 2015 [8]. It contains 728 sub-links (364 links, each measured in both directions) sampled every 10 s, with the RSL quantised to 0.3 dB and the TSL to 1 dB, at 7.5 to 38.7 GHz. A weather radar gives one rain map every 5 min on a grid of about 2 km, and ten city gauges measure the rain on the ground every minute. A data pipeline cleaned and synchronised the links, averaged them to 1 min and 5 min values, and marked rain events and quality problems. We kept the 28, 32 and 38 GHz bands, which hold 660 of the 728 sub-links (91 %). The other 68 sub-links belong to links of 4 km or more, so their antenna part cannot be separated from their path part by comparing lengths.

2.2. Split and selection

Whole rain events were assigned before any modelling to a training set (TRAIN) for fitting, a validation set (VAL) for choosing between model versions, and a test set (TEST) for one final check. In addition 71 of the 364 links (20 %) were held out as unseen links. A link-event, one sub-link during one event, was kept only if it had a baseline, rain on the link, a good sub-link status, a stable offset (spread at most 0.3 dB) and no TSL step during the rain (Table I). Rain is rare. Of the TRAIN minutes, 12 % have a rain rate of 0.5 mm/h or more and 0.4 % have 10 mm/h or more, so every result above 10 mm/h rests on few data.

2.3. The rain input

The model needs the rain along every path every minute. Only the radar covers every path, but its scale error differs from storm to storm, while the gauges are accurate but measure at ten points. The two are combined: the radar gives where and when it rains on a path, and the gauges give how much. For each event in which the gauges caught at least 1 mm per gauge link, the radar is divided by one factor per event,

$$R(t) = \frac{R_{\mathrm{radar}}(t)}{g_e}, \qquad g_e = \frac{\text{radar rain in event } e}{\text{gauge rain in event } e}, \quad (3)$$

and $g_e = 1$ otherwise. Pooled over the events, the radar measured 0.89 times the gauge rain on TRAIN and 1.71 times on VAL. Section 4.3 shows why this factor matters.

2.4. The baseline

For each event and sub-link the baseline is one constant, the median attenuation over the 5 min slots of the 24 h before the event that radar and gauges mark as dry and that start at least 1 h after the last rain. On fake events placed in dry periods, where the true WAA is zero, the baseline error had a median of 0.00 dB and 10th and 90th percentiles of −0.21 and +0.22 dB, and one hour after real rain the baseline was only 0.04 dB above that of a dry control (Appendix A). The offset c is the median of A over the dry minutes 120 to 30 min before the first rain on the link.

TABLE I. The split of the data. A link-event is one sub-link during one rain event. Two of the 53 TRAIN events hold no usable link-event. The largest loss is the link-events without rain on the link.

Set	Events	Link-events	Use
TRAIN	53	9 780	fitting (206 links)
VAL	18	3 250	choosing between versions
TEST	18	3 355 + 1 010	one check (+ 59 new links)

TABLE II. The assumptions of Eq. (4).

Assumption	Physical reason	Tested in
The two parts add	separate losses in series	length test, Sec. 3.2
W does not depend on L	two antennas on every link	Sec. 4.3
Path part is L times a loss per km	equal rain, equal loss per km	quadratic term in 4 of 15 classes
R_{radar}/g_e is the rain	radar for where, gauges for how much	VAL gap closed, Sec. 4.3
Rain drives the wetting	the water comes from rain	dew excluded, Sec. 5

3. METHOD

3.1. The decomposition

During an event the measured attenuation of a link is written as

$$A(t) - c = W(t) + L\,P\big(R(t-d)\big), \tag{4}$$

where $P(R)$ is the specific path attenuation in dB/km and d a lag of whole minutes. Neither W nor P has a form imposed in advance. Table II lists the assumptions, the physical reason for each, and where each is tested. The path term $L\,P(R)$ uses the mean rain along the path. It is exact for any rain pattern when P is linear in R, because then $\int c\,R(x)\,dx = c\,L\,\bar{R}$, and the data show that P is close to linear (Sec. 4.2).

3.2. The length test

For one link, Eq. (4) has two unknowns. The two parts, however, depend differently on the length. The antenna part is the same on every link, while the path part grows with L. Between links of different length in the same rain, a difference in $A - c$ therefore comes from the path. In practice we collect the 5 min values of steady rain whose path rain falls in one class (0.5–1, 1–2, 2–5, 5–10 and 10–20 mm/h) from all links and events, take for each link and event the median of $A - c$ in that class, and fit a robust (Huber) straight line

$$A - c = I + s\,L. \tag{5}$$

The intercept at $L = 0$ is the antenna part, $I = W$, and the slope is the loss per km, $s = P$. The 90 % intervals come from resampling whole events (1000 draws). Each band and rain class (a cell) gives one point of the curves $W(R)$ and $P(R)$, and a cell is used for a verdict only if it holds at least 8 links shorter than 1 km. No model enters this test. The same line can also be fitted through a model's own predictions on the same 5 min values, which tells whether the model splits A the way the data do.

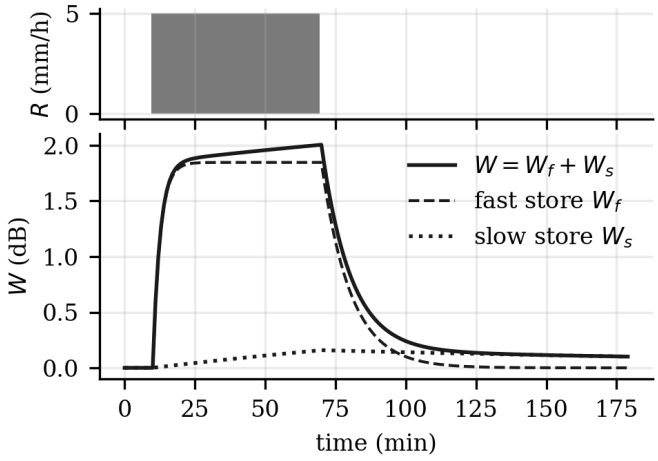


FIG. 2. Response of the final equation (computed, not measured) to 60 min of 5 mm/h on a 28 GHz link with $a_{\mathrm{link}} = 1$. The fast store fills within minutes and empties within about 30 min, and the slow store remains for hours.

3.3. Memory: a two-store model

The water on a radome does not follow the rain instantly. We model it as a store that relaxes toward a target set by the current rain,

$$\frac{dW}{dt} = \frac{u(t) - W}{\tau}, \qquad u(t) = a_{\mathrm{link}}\,F\big(R(t-d)\big), \tag{6}$$

where $F(R)$ is the steady WAA in constant rain R and a_{link} a dimensionless factor per link (the link factor), because some antennas collect more water than others. When W is below its target the store fills (wetting), and when it is above, it empties (drying). Wetting and drying have their own time constants, and a small part of the water stays for hours, so the store is split into a fast part with a share φ and a slow part,

$$\frac{dW_f}{dt} = \frac{\varphi\,u - W_f}{\tau_f}, \qquad \tau_f = \begin{cases} \tau_{\mathrm{wet}} & \text{if } \varphi\,u > W_f \\ \tau_{\mathrm{dry}} & \text{otherwise,} \end{cases} \tag{7}$$

$$\frac{dW_s}{dt} = \frac{(1-\varphi)\,u - W_s}{\tau_s}, \qquad W = W_f + W_s. \tag{8}$$

For a step in the rain the fast store follows $\varphi u\,(1 - e^{-t/\tau_{\mathrm{wet}}})$ while wetting and $W_0\,e^{-t/\tau_{\mathrm{dry}}}$ while drying, with W_0 the water left when the rain stops. Figure 2 shows the response of the final equation to one hour of steady rain.

The radar, with its 5 min maps, cannot resolve time constants of a few minutes. On fake data a 5 min noisy input alone made the fitted wetting 2.1 times slower. The time constants and the share were therefore measured with the gauges, which record the start and end of rain to the minute, and then held fixed in the radar fit.

3.4. Learning F and P, then choosing closed forms

All parts of Eqs. (4)–(8) were fitted together on TRAIN with F and P as free curves, that is, values at fixed rain rates from 0.05 to 40 mm/h joined by straight lines. The loss is a Huber loss of the 5 min residuals with a 0.3 dB threshold, and $\ln a_{\mathrm{link}}$ carries a weak prior around zero, which fixes the product $a_{\mathrm{link}}F$ but not the level of F alone. Only after the free curves were known were closed forms fitted to them. Physical forms (curves that saturate at a maximum, capped power laws) and forms proposed by symbolic regression with PySR [9] competed under one rule, the corrected Akaike information criterion (AICc), which weighs the fit against the number of parameters. The chosen form was put back into the full model, refitted, and accepted only if it predicted unseen events as well as the free curve and still passed the length test.

3.5. Validation protocol

Every step followed the same protocol. The pass rule of a check was written down, dated, before the check was run. Every fitted number was first recovered on fake data generated from the model with realistic noise on the real links and the real rain. Intervals come from refitting on resampled sets of whole events. Structural choices were judged on VAL, never on TRAIN alone, so numbers on VAL are optimistic. Before TEST was read, the equation, three simpler comparison models and the pass rules were frozen and their cryptographic hashes committed, and TEST was then read once. A decision that deviated from a written rule was dated and its reason recorded (Appendix B).

4. RESULTS

4.1. What the data say without a model

Figure 3 shows the length test on TRAIN. In each rain class the points of $A - c$ lie around a straight line in L. The intercepts, the antenna part, rise from about 1.0 dB at 0.5 to 1 mm/h to 2.4 to 2.8 dB at 5 to 20 mm/h and then grow only slowly, while the slopes, the path loss per km, grow roughly in proportion to R. A WAA that is constant in rain would give a flat intercept, which the data exclude, and a WAA that grows with the path would show up as a slope that does not follow the rain. The test also showed why short links alone are not enough. When all links are short, a small path loss and a slightly larger antenna loss have nearly the same effect, and the fitted path loss on links below 1 km halved when links up to 3 km were added. The model was therefore fitted on all lengths together.

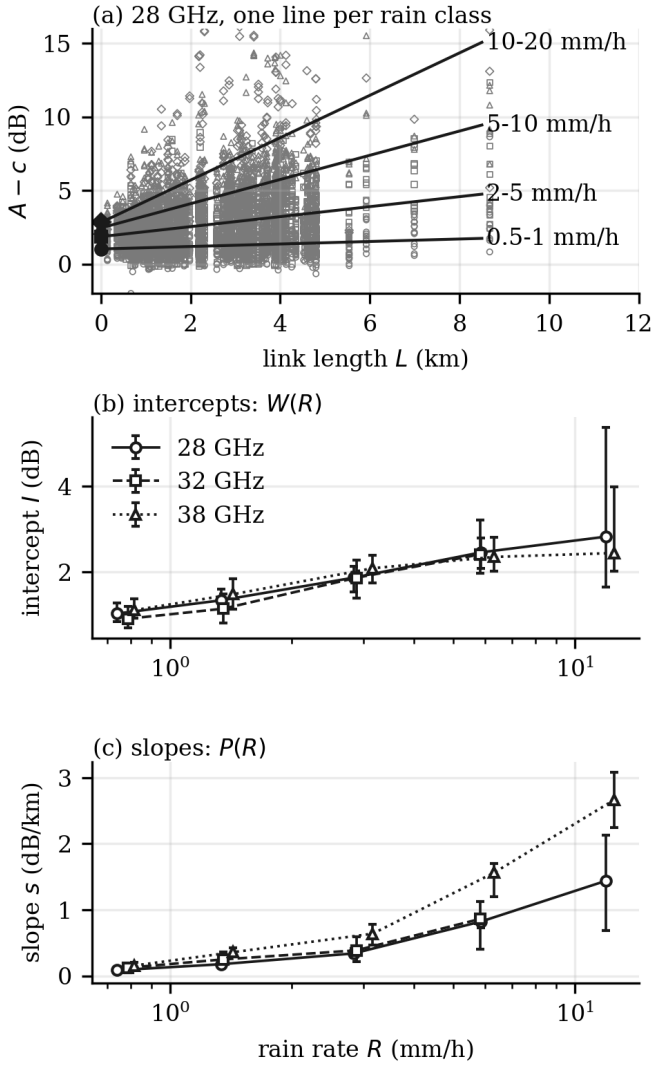


FIG. 3. The model-free length test on TRAIN. (a) Each point is one link in one event at 28 GHz, grouped by rain class (marker shape), with the robust line of each class. (b) The intercepts at $L = 0$ give the antenna part $W(R)$. (c) The slopes give the path attenuation per km $P(R)$. Bars are 90 % event-bootstrap intervals. Line style and marker show the band.

4.2. The equation

On the free curves, one shape of F scaled by one level per band predicted VAL as well as separate shapes per band (RMSE 0.3875 against 0.3878 dB), although TRAIN alone favoured separate shapes. F rises steeply in light rain and more slowly above, and P is close to linear in R (Fig. 5). The simplest closed form that describes F as well as the free curve is a power of R close to a square root, proposed by PySR, and the linear form was chosen for P. The full equation with all its numbers is given in Fig. 4. The exponent $b = 0.54$ has the interval 0.47 to 0.63, which contains 0.5 and excludes a linear dependence. The formula has no maximum, so the data do not show

whether WAA saturates, and it is not valid above 40 mm/h.

All nine numbers of the closed form were recovered within 20 % on fake data, as were all 62 numbers of the free curves, 61 of them within 10 %. Eight numbers, the path loss below 1 mm/h, are too close to zero to be distinguished from it and are not reported. The data fix the product $a_{\mathrm{link}}F$. The level of F alone is set by the weak prior that keeps the median a_{link} near 1.

4.3. WAA does not depend on the link length

The central assumption of Eq. (4) was tested in three ways that need no rain law, before TEST was read. First, the link factor does not depend on the length (Fig. 6). Its Spearman correlation with L is +0.05 (interval −0.07 to +0.17) for the model frozen for the test and +0.09 (−0.03 to +0.20) for the final equation. Second, the length test repeated with links shorter than 3 km only and with links longer than 1.5 km only gave the same intercept in 13 of 15 cells. Third, in 9 081 periods after rain, on 206 links in 23 TRAIN and 10 VAL events, the radar showed no rain on any pixel of the path. There the path term is zero, so $A - c$ is the WAA alone, measured without any model. The measured WAA and the model error were the same for short, medium and long links, with a median error of −0.18 dB at 10 min in every length class (Fig. 7).

The same tests located the one large gap between TRAIN and VAL in the rain input, not in the antennas. With the raw radar the VAL length test agreed in at most 1 of 8 cells, and with the gauge factor of Eq. (3) in 5 of 6. A rate-dependent radar error, which would need a curve instead of one factor, was not found: its apparent dependence reversed when the data were binned by the radar instead of the gauge, the signature of regression to the mean between two noisy measurements.

4.4. Comparison with other WAA models

All models were compared on the same task. Each receives only the corrected rain R, the length and the band, and predicts the 5 min attenuation $A - c$ with the same events, baselines, offsets and path term $L\,c_{\mathrm{band}}R$, and each was fitted on TRAIN only. Three comparison models were frozen with our equation before TEST was read: no WAA, a constant WAA while it rains, and a WAA without memory that has the same level curve as ours but no stores. The two published models were added after TEST had been read, without any change to our equation, so their comparison is post-freeze. The Pastorek model [5] is $W = C\,(1 - e^{-0.1R^{\zeta}})$, with an upper limit C per band and no memory. Its ζ acts inside a saturating function and is not comparable with our b. The Schleiss model [4] rises toward $W_{\max}$ while it rains and decays exponentially after, in the discrete form $W \leftarrow W + (W_{\max} - W)\,3\Delta t/\tau_{\mathrm{wet}}$ of the open-source package pycomlink (version 0.6.0). That code uses the attenuation measured on the link itself, which would give the model the answer, so a minute is wet here when $R > 0.1$ mm/h, and the dry step decays with its own τ_{dry}. Each published model was used in two versions. Version (a) keeps the default parameters and fits only c_{band}. For the Schleiss model this requires a drying time that could not be checked in the paper, and it was set equal to the wetting time (15 min), so version (a) is the pycomlink defaults with an assumed drying time, not the published model. Version (b) fits all parameters on TRAIN with the same link factor as our equation, and its verdict is the one that counts.

Table III and Fig. 8 give the accuracy. Our equation is better than every model whose interval lies above zero, which is all of them except the refitted Schleiss model, with which it ties, on all links and on links below 1 km. Accuracy alone, however, does not show that a model splits A correctly into antenna and path. A model can trade one part for the other and still fit the total. Table IV therefore compares each model with four measurements that are independent of its fit: the slope and intercept of the length test, the WAA measured on dry paths after rain, and the gauge timing. It also gives the correlation of each model's link factor with L, which should be zero if the model's WAA does not depend on the length. This last check was added after the other numbers of Table IV had been seen, because the refitted Pastorek model has a path loss of only 0.6 to 0.7 times ours yet the measured slope. No pass rule was written for this table, and its numbers are reported as they came out.

The conclusion can be stated precisely. In accuracy our equation is better than or equal to every model, and it ties only with the refitted Schleiss model. Of the models with a link factor, it is the only one whose link factor is independent of the length. Its path loss agrees with the measured slope in 7 of 8 cells (median ratio 1.04). After rain no model agrees with the measured WAA within its interval. Ours holds 0.11 to 0.19 dB too much water, the refitted Schleiss model 0.18 to 0.25 dB too much, and the models without memory 0.15 to 0.35 dB too little (Fig. 9). Only the default Schleiss version, with its assumed drying, has a smaller error at 5 and 10 min, at the price of a slope inside the measured interval in only 5 of 8 cells. Our timing agrees with the gauges by construction, since it was measured with them, so it is not independent evidence. Our equation is not the best on every measure. Its intercept agrees in only 4 of 8 cells, fewer than the refitted Schleiss (7), the refitted Pastorek (6) and the model without memory (6), which is the failure discussed in Sec. 4.5. The refitted Schleiss model, which ties in accuracy, reaches its fit with a path loss 1.7 to 1.8 times ours, a slope too high in heavy rain, a wetting time of about 100 min and a drying time of about 25 min against the gauge times of 2.6 and 10.3 min, more water than measured after rain, and a link factor that falls with the length. The refitted Pastorek model has a link factor that rises with the length and no water at all after rain. The published models have neither fake-data

MEASURED QUANTITY. $A(t) = \mathrm{TSL}(t) - \mathrm{RSL}(t) - B$ in dB at 1 min, B the median dry attenuation of the 24 h before the event (48 h if less than 1 h of dry data remains), c the median of A over the dry minutes 120 to 30 min before the first rain on the link (0 if fewer than 30 such minutes). INPUTS. Link length L (km), band from the frequency, 28 GHz (25–30.5 GHz), 32 GHz (30.5–35 GHz), 38 GHz (35–45 GHz), link factor a_{link}, and the rain $R(t) = R_{\mathrm{radar}}/g_e$ in mm/h, one value per minute, with R_{radar} the radar rain averaged over the pixels along the path and held over its 5 min interval and g_e the ratio of radar to city-gauge rain of the event over the gauges within 1 km of a path ($g_e = 1$ if the gauges caught less than 1 mm per gauge link).

$$A(t) - c = W_f(t) + W_s(t) + L\, c_{\mathrm{band}}\, R(t-d), \qquad u(t) = a_{\mathrm{link}}\, A_{\mathrm{band}} \left(\frac{R(t-d)}{5\ \mathrm{mm/h}} \right)^{b},$$

$$\frac{dW_f}{dt} = \frac{\varphi\, u(t) - W_f}{\tau_f}, \quad \tau_f = \begin{cases} \tau_{\mathrm{wet}} & \text{if } \varphi\, u > W_f \\ \tau_{\mathrm{dry}} & \text{otherwise} \end{cases}, \qquad \frac{dW_s}{dt} = \frac{(1-\varphi)\, u(t) - W_s}{\tau_s}.$$

Initial state $W_f = W_s = 0$ in dry weather before the event. One step per minute, $W \leftarrow v + (W - v)\, e^{-1/\tau}$ with v the target of the store, output after the step.

Parameter	Value (90 % interval)	Source
A_{28}, A_{32}, A_{38} (dB)	2.57 [2.00, 3.21], 2.36 [1.70, 3.06], 2.77 [2.25, 3.40]	fitted on TRAIN, 40 event-bootstrap refits
b	0.54 [0.47, 0.63]	fitted, as above
c_{28}, c_{32}, c_{38} (dB km^{-1} per mm/h)	0.105 [0.058, 0.138], 0.141 [0.047, 0.190], 0.194 [0.111, 0.245]	fitted, as above
a_{link}	one per TRAIN link (206 links), 10th to 90th percentile 0.77 to 1.24, 1 for a new link	fitted with the above
τ_{wet}, τ_{dry} (min)	2.6 [2.4, 3.1], 10.3 [9.2, 11.8] (VAL 2.0 [1.5, 3.2], 8.8 [5.8, 13.1])	fixed, from the gauge-input fit on TRAIN
φ	0.72 [0.60, 0.81]	fixed, as above
τ_s (min)	240	fixed at the search bound, not identified
d (min)	1	selected in the fit

VALIDITY. $0 \le R \le 40$ mm/h (above 10 mm/h based on 0.4 % of the minutes), the three bands, link lengths 0.14 to 8.7 km, OpenMRG (Gothenburg, summer 2015). Not identified: τ_s and φ, the path loss below 1 mm/h, and the level of F apart from a_{link}.

FIG. 4. The WAA equation of Part 1, frozen for the test, with the closed form of F adopted after it (Sec. 4.5). All values and the per-link factors are stored in the equation file of the project and computed by its reference implementation, which reproduces the fit to 10^{-15} dB.

TABLE III. Accuracy on TEST: RMSE of the 5 min attenuation in dB, and the paired difference to our equation (model minus ours, 90 % interval from 1000 resampled sets of storms, the same draws for every model). Above zero means ours is better. Ours: 0.695 [0.514, 0.803] dB for all links and 0.564 [0.436, 0.633] dB below 1 km.

Model	RMSE	Minus ours, all	Minus ours, $L < 1$ km
No WAA	0.862	+0.167 [+0.095, +0.210]	+0.275 [+0.131, +0.357]
Constant WAA	0.760	+0.065 [+0.047, +0.075]	+0.080 [+0.047, +0.099]
No memory	0.727	+0.032 [+0.019, +0.049]	+0.037 [+0.021, +0.058]
Schleiss (a)	0.859	+0.164 [+0.084, +0.286]	+0.200 [+0.111, +0.329]
Schleiss (b)	0.703	+0.009 [-0.010, +0.039]	+0.012 [-0.009, +0.044]
Pastorek (a)	0.725	+0.031 [+0.017, +0.047]	+0.027 [+0.005, +0.060]
Pastorek (b)	0.720	+0.026 [+0.015, +0.038]	+0.031 [+0.017, +0.046]

recovery nor intervals.

4.5. The test rules and the light-rain level

TEST contains 18 storms, with 3 355 link-events on known links and 1 010 on 59 links that the model had never seen. After the read, two errors were found in the evaluation code, not in the model: one storm with a radar-to-gauge ratio of zero and three storms without a ratio had an undefined rain input. Both received the fallback $g_e = 1$ that the rule already specified, and the numbers before and after the fix were recorded. Table V gives the five rules, written before the read, for the model frozen for the test.

The failed rule locates the limit of the equation. The slopes, which measure the rain on the path, agree in 7 of 8 cells. The intercepts, which measure the WAA, disagree

TABLE IV. Physical consistency on TEST (post-freeze summary, nothing refitted). Length test: of the 8 eligible cells, the number in which the model's own slope s or intercept I lies inside the model-free 90 % interval, and the median ratio model / measured. Dry path: measured minus model WAA after the path became dry (1791 windows, median, positive means the model holds too little water). The measured WAA is 0.35, 0.27 and 0.15 dB at 5, 10 and 20 min. $\rho(a_{\mathrm{link}}, L)$: Spearman correlation over the TRAIN links, 90 % link-bootstrap interval. Times: 1/e times of wetting and drying in min (Schleiss: of its discrete step), against the gauge timing 2.6 and 10.3 min (VAL 2.0 and 8.8 min). Parameters: fitted global numbers, with a_{link} per link counted apart.

Model	s in	s ratio	I in	I ratio	Dry path 5/10/20 min (dB)	$\rho(a_{\mathrm{link}}, L)$	Wet/dry (min)	Param.	Identified
No WAA	0	2.36	0	0.01	+0.35 / +0.27 / +0.15	–	–	3	no check
Constant WAA	4	1.32	0	0.54	+0.35 / +0.27 / +0.15	−0.60 [-0.66, -0.52]	0 / 0	6	no check
No memory	7	0.97	6	0.81	+0.35 / +0.27 / +0.15	+0.20 [+0.08, +0.32]	0 / 0	12	no check
Schleiss (a)	5	0.65	2	1.38	−0.05 / +0.10 / +0.13	–	4.5 / 4.5	3	no check
Schleiss (b)	6	1.15	7	0.78	−0.24 / −0.25 / −0.18	−0.45 [-0.55, -0.35]	103 / 25	8	no check
Pastorek (a)	4	0.58	7	1.05	+0.33 / +0.25 / +0.15	–	0 / 0	3	no check
Pastorek (b)	7	0.97	6	0.82	+0.33 / +0.25 / +0.15	+0.31 [+0.18, +0.42]	0 / 0	7	no check
Ours	7	1.04	4	0.72	−0.19 / −0.15 / −0.11	+0.09 [-0.03, +0.20]	2.6 / 10.3	7	9 of 9 (fake data)

TABLE V. The five test rules, written before TEST was read, and the results for the frozen model.

Rule	Result	Verdict
P1 length test	4 of 8 cells agree (6 needed): slopes 7 of 8, intercepts 4 of 8	fail
P2 beats simpler models	0.692 dB against 0.862, 0.760, 0.727; below 1 km 0.562 against 0.838, 0.644, 0.601	pass
P3 equal WAA by length	dry-path error, long minus short, −0.04 [-0.15, +0.04] dB	pass
P4 no class out of line	TEST / VAL ratio 1.12, each class within 5 % of it	pass
P5 new links	error 1.07 times that on known links (limit 1.20)	pass

in light rain (0.5 to 2 mm/h), where the test storms had 1.35 to 1.55 dB against 0.83 to 1.11 dB in the equation. The WAA level in light rain therefore varies between storms more than the equation describes. The failure is not a dependence on length, which P3 tested and passed.

The equation frozen for the test described F by its free curve. The closed form had been fitted before the freeze but was not the frozen version. After the test it was evaluated on the same data without refitting, under a rule written before the evaluation: adopted only if it gives the same five verdicts and its error lies within ±0.01 dB of the free curve. Both held, with the same verdicts and a difference of +0.0025 dB (−0.0065 to +0.0085), so the closed form is the written form of F (Fig. 4).

5. DISCUSSION

The tie with the refitted Schleiss model is not an equivalence. Two models can predict the total attenuation equally well while they split it differently between the antennas and the path, because during most of a storm both parts rise and fall with the same rain. The split is visible only where the two parts behave differently, along the length and after the rain, and there the two models disagree. The Schleiss fit puts 1.7 to 1.8 times more loss on the path and compensates on short links with a link factor that falls with length. In effect its WAA depends on the length, which our three tests of Sec. 4.3 do not find in the data. Its wetting time of about 100 min also contradicts the gauge timing, which shows the film forming within 2 to 3 min. Equal accuracy with an unphysical split would carry the error into Part 2, where the path law is learned after the WAA is removed.

The one failed test rule concerns the WAA level in light rain. On the test storms the intercept at 0.5 to 2 mm/h was about 0.4 to 0.6 dB higher than the final equation, while the slope agreed. A single curve $F(R)$ cannot carry a level that depends on the storm, for example on how long it rained before or on the drop sizes of light rain, but neither cause was tested. The refitted Schleiss and Pastorek models and the model without memory match this level more often (6 or 7 of 8 intercepts), but they do so with a link factor that changes with the length or with no water after the rain, so their better intercept does not come with a better split. Why our equation sits low in light rain is not known. The slow store, which is not identified, is one candidate, because it sets how much of the level is reached after a given time of steady rain.

After the rain our equation holds about 0.25 dB more water than measured at the moment the path becomes dry, and about 0.5 dB after heavy rain (Fig. 7). The causes were tested one by one. The measured decay rate is 0.87 times that of the model (interval 0.70 to 1.06), so the drying speed is consistent, and the error lies in the amount of water at the end of the rain. It is not caused by the radar timing, and the model is not too high during heavy rain. The error is larger after heavier rain and in drier air. Water running off the radome, which has been seen in the laboratory [7], would explain this, but this work cannot confirm it. A version with a free drying time of 3.7 min removes the misfit, but it was

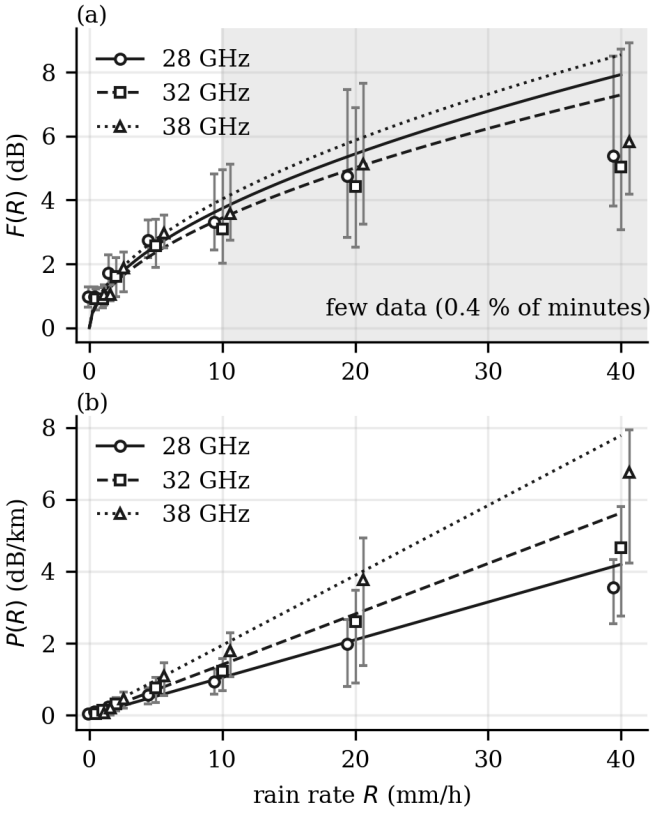


FIG. 5. The learned curves on TRAIN. Points: free curves with 90 % intervals. Lines: the closed forms, one line style per band. (a) Steady WAA $F(R)$ and $A_{\mathrm{band}}(R/5)^{0.54}$. Above 10 mm/h (shaded) F is poorly determined. (b) Path attenuation per km $P(R)$ and $c_{\mathrm{band}}R$.

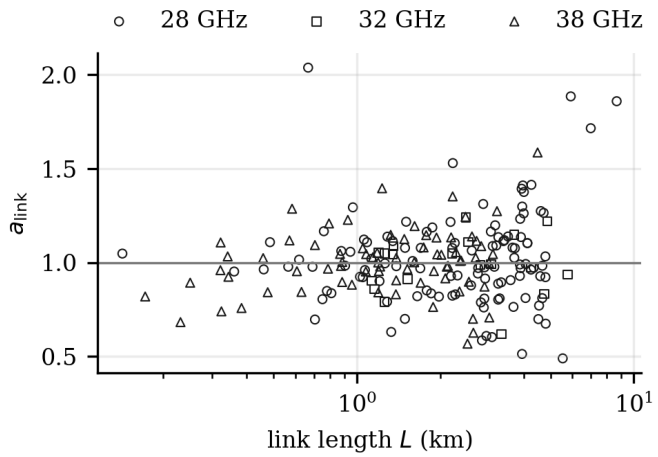


FIG. 6. The link factor a_{link} of each of the 206 TRAIN links against its length. Marker shape gives the band. There is no trend.

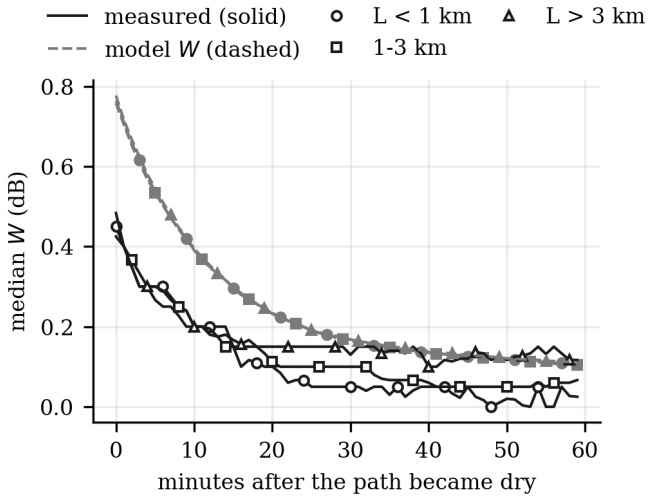


FIG. 7. WAA measured after rain where no rain remains on the path (solid black) against the model (dashed grey), median over 9 081 windows of TRAIN and VAL, by length class (marker). The three classes agree. The model starts too high (Sec. 5).

not adopted. With so fast a drying the antenna part follows the rain almost as quickly as the path part, the two can no longer be separated, and the path loss halves with an interval reaching zero in every band. The 3.7 min is an effective time for the 5 min radar input, not the drying time measured with the gauges. The faster drying therefore stays outside the equation, and so does the wind, whose apparent effect on the drying (Spearman −0.76 over 17 storms) disappeared once the preceding rain was held fixed.

Dew wets the same antennas without rain. On 125 humid link-nights the attenuation rose by a median of 1.7 dB for about 4 h, and links with a larger a_{link} also had more dew (Spearman +0.26). Dew is therefore a real WAA, but it is not driven by rain and was left out of the equation on purpose.

The error budget shows where the remaining error lies. On TEST, 61 % of the squared error occurs during rain within storms, 8 % is the difference between storms (the rain reference), 6 % falls in the hour after rain and 25 % in dry minutes. On VAL the shares within storms and after rain are 65 % and 2 %. The after-rain misfit is thus a small part of the error. A likely cause of the large part is that the radar resolves rain more coarsely (5 min, 2 km) than a link does, but this has not been tested.

The path loss of the equation is linear in R, $P = c_{\mathrm{band}}R$. A power law fitted to the free points of P gave an exponent of 0.97 with a quadratic term of zero, and the ITU-R P.838 exponents at 28 to 38 GHz are 0.86 to 0.92 [2], so both describe a nearly linear loss in this range. Its size is 0.52 to 0.74 times the ITU-R value. Whether this is a remaining scale error of the radar rain or an effect of uneven rain along the path, which is what $f(L, \bar{R}, Z)$ in Part 2 is meant to capture, cannot be decided here.

Table VI states what the model claims. Box A is in the equation, with identified numbers. Box B is identified but kept outside, because adding it would unbind another number. Box C is seen but not explained.

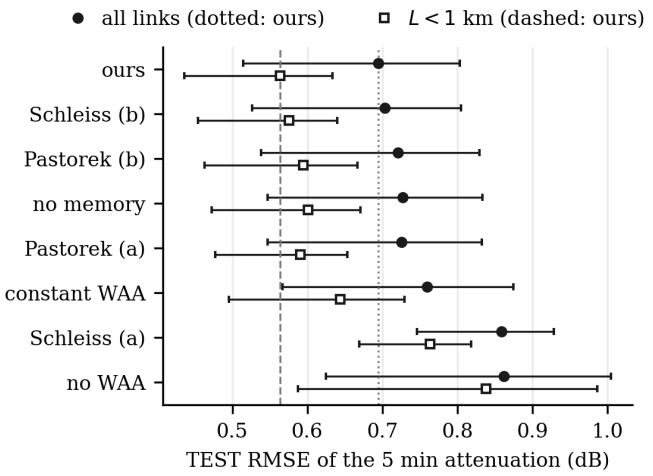


FIG. 8. TEST RMSE of every model, all links (filled circles) and links below 1 km (open squares), with 90 % intervals. The vertical lines mark our equation.

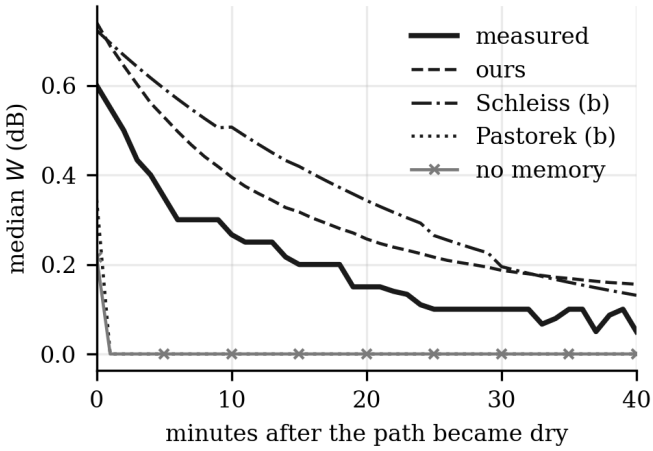


FIG. 9. WAA on TEST after the path became dry: measured (thick solid) against our equation and the refitted published models, median over 1791 windows on all lengths. The model without memory (grey crosses) and the Pastorek model fall to zero when the rain stops.

6. CONCLUSIONS

A dynamic wet antenna law was identified from the OpenMRG links without subtracting any rain law. The antenna and the path were separated by the link length and by the dry paths after rain, and the result is a fast and a slow store whose target grows about as the square root of the rain rate, with a wetting time of 2.6 min, a drying time of 10.3 min, and a path loss linear in the rain. On held-out storms, read once after a hashed freeze, it predicts the attenuation better than or as well as every other model tested, and of the models with a link factor it is the only one whose link factor does not change with the link length. Its limit is the WAA level in light rain, which varies between storms. The next step is to remove the WAA with this equation and test whether the rain estimated from the links agrees better with the gauges, which needs its own written plan and a decision about TEST, which has already been read once. Part 2 will then learn the path law $f(L, \bar{R}, Z)$ with the WAA removed by the frozen equation.

TABLE VI. What the model claims.

Box	Content
A, in the equation	the level rising about as $\sqrt{R}$, one shape for all bands, the fast store (2.6 and 10.3 min), the slow store as a structure, $P = c_{\text{band}} R$, the band levels, the link factor, W independent of L
B, outside	faster drying with wind, the end-of-rain level (larger after heavy rain and in dry air), dew
C, unexplained	the numbers of the slow store, the cause of the link factor, F above 10 mm/h, P below 1 mm/h, the WAA level in light rain between storms

Appendix A: The baseline

For each event and sub-link the baseline is one constant. The 24 h before the event start are taken (48 h if less than 1 h of dry data remains), only the 5 min slots that radar and gauges mark as dry and that start at least 1 h after the last rain are kept, and the baseline is the median attenuation over them. The rule of 1 h excludes the drying after earlier rain. Links with a large daily cycle of the dry level or with erratic behaviour are flagged and excluded. The method was tested where the truth is known by placing fake events inside dry periods, where the correct WAA is zero. For 6 h events the median error was 0.00 dB, with 10th and 90th percentiles of −0.21 and +0.22 dB, so the error has no bias and averages out over many events. Each sub-link keeps its own measured error, and sub-links with an error above 0.3 dB are excluded. A second test checked that the baseline does not absorb the drying after rain. One hour after real rain the level was 0.04 dB above that of a dry control, and the difference fell to zero within 12 h. The same test showed the rises on humid nights that are attributed to dew.

Appendix B: Decisions

Table VII lists the decisions that changed the course of the work, with their dates and reasons. Decisions marked as deviations override a written rule, and each is stated openly.

TABLE VII. The dated decisions of Part 1.

Date (2026)	Decision	Reason
7 Oct	Baseline frozen (median of the dry 24 h)	unbiased on fake events, does not absorb the drying
13 Oct	Drying tails closed, restart as system identification of whole events	tails wait or rise, most of a tail passes while it rains nearby, no law identified (Appendix C)
17 Oct	Radar divided by the event's radar-to-gauge factor	VAL length test 1 of 8 cells with the raw radar, 5 of 6 with the factor
18 Oct	Two-store model with gauge timing kept, a more flexible slow store dropped (deviation)	its slow store was not identified, gain 0.0005 dB on VAL; decided after seeing VAL
19 Oct	Closed form of F not adopted before the test	failed the length test at exactly 4 of 6 cells against a written 0.6667
19 Oct	Faster drying (3.7 min) kept outside the equation (deviation)	passed its rules but unbound the path loss
19 Oct	Freeze, hashes committed, TEST read once	protocol
19 Oct	Closed form of F adopted after the test	same five verdicts, difference within ±0.01 dB, rule written first
28 Oct	Schleiss (b) searched on a coarse grid, then refined	first run stopped by low memory, changed before any result
2 Nov	Link factor against L added to Table IV	added after its other numbers were seen

Appendix C: Side roads

The first phase studied drying tails, the stretches of A after rain. From 20 355 extracted tails a frozen set of 429 training tails was selected by rules that never looked at the decay. A simulator calibrated to the real tails showed that fitting tails one at a time cannot identify a drying law, that fitting all tails together with a shared law can, and that SINDy [10] applied directly to the noisy, quantised tails cannot. On the real tails no drying law was identified, because about one tail in five stays flat or rises after the rain, most of a tail passes while it still rains nearby, and the fast first drop does not grow with the path length. The tail was the wrong object, and the phase was closed on 13 October.

The system identification that replaced it fitted a ladder of models of growing complexity to whole events, each first checked on fake data. A flexible rung in which the wetting and drying could be any function of rain and stored water, to be named afterwards by sparse regression or PySR, was not identified on real data. Factors per event, fitted to absorb the differences between storms, failed their fake check on light events. A slow store with separate wetting and drying times passed its rule but its numbers were not identified (Appendix B). A wind-dependent drying improved the fit but failed the rules that require stable numbers, and a drying that depends on the preceding rain had no effect. A rain-rate dependent correction of the radar was not justified. Pairs of links on a common path, which would measure W directly, were too few to test (4 pairs).

Appendix D: Uncertainty and identifiability

All fits minimise a Huber loss of the 5 min residuals (threshold 0.3 dB), which limits the effect of outliers, plus a prior on $\ln a_{\mathrm{link}}$. Intervals of the equation's numbers are the 5th and 95th percentiles of 40 refits on sets of TRAIN events drawn with replacement, each refit started cold. Resampling whole events keeps the correlation between the minutes and the links of one storm. Profile intervals, which vary one number and refit the others, were 1 to 13 times narrower than the bootstrap intervals for most numbers, so the event-to-event variation of the rain reference dominates the uncertainty. A number counts as identified if it is recovered within 20 % on three fake datasets generated from the fitted model with realistic noise on the real links and rain, and if its interval does not reach a search bound. Test intervals are the 5th and 95th percentiles of 1000 resampled sets of TEST storms, with the same draws for every model so that differences are paired. The fixed time constants come from warm-started refits of the gauge-input fit, and their intervals may be too narrow.

Appendix E: Full TEST tables

Table VIII gives the TEST error by length class. Table IX gives the eight cells of the length test with the measured intervals and the values of each model. The full tables, including the links never fitted, are in the step reports of the project.

[1] H. Messer, A. Zinevich, and P. Alpert, "Environmental monitoring by wireless communication networks," *Science* **312**, 713 (2006). https://doi.org/10.1126/science.1120034

TABLE VIII. TEST RMSE (dB) by length class. 90 % intervals are in the step reports.

Model	$L < 1$ km	1–3 km	$L > 3$ km	All
Ours	0.564	0.684	0.780	0.695
No WAA	0.838	0.850	0.895	0.862
Constant WAA	0.644	0.750	0.837	0.760
No memory	0.601	0.715	0.812	0.727
Schleiss (a)	0.763	0.845	0.934	0.859
Schleiss (b)	0.576	0.693	0.787	0.703
Pastorek (a)	0.590	0.706	0.826	0.725
Pastorek (b)	0.595	0.709	0.805	0.720

[2] ITU-R, "Specific attenuation model for rain for use in prediction methods," Recommendation ITU-R P.838-3 (2005).

[3] H. Leijnse, R. Uijlenhoet, and J. N. M. Stricker, "Microwave link rainfall estimation: effects of link length and frequency, temporal sampling, power resolution, and wet antenna attenuation," *Adv. Water Resour.* **31**, 1481–1493 (2008). https://doi.org/10.1016/j.advwatres.2008.03.004

[4] M. Schleiss, J. Rieckermann, and A. Berne, "Quantification and modeling of wet-antenna attenuation for commercial microwave links," *IEEE Geosci. Remote Sens. Lett.* **10**, 1195–1199 (2013). https://doi.org/10.1109/LGRS.2012.2236074

[5] J. Pastorek, M. Fencl, J. Rieckermann, and V. Bareš, "Precipitation estimates from commercial microwave links: practical approaches to wet-antenna correction," *IEEE Trans. Geosci. Remote Sens.* **60**, 1–9 (2022). https://doi.org/10.1109/TGRS.2021.3110004

[6] J. Tiede, C. Chwala, and U. Siart, "New insights into the dynamics of wet antenna attenuation based on in situ estimations provided by the dedicated field experiment ATTRRA2," *IEEE Geosci. Remote Sens. Lett.* **20** (2023). https://doi.org/10.1109/LGRS.2023.3320755

[7] J. Tiede, C. Chwala, U. Siart, and T. Eibert, "Frequency-dependent variations of the antenna reflection coefficient due to different wetness conditions on the antenna radome," *Adv. Radio Sci.* **23**, 13–19 (2025). https://doi.org/10.5194/ars-23-13-2025

[8] J. C. M. Andersson, J. Olsson, R. van de Beek, and J. Hansryd, "OpenMRG: Open data from Microwave links, Radar, and Gauges for rainfall quantification in Gothenburg, Sweden," *Earth Syst. Sci. Data* **14**, 5411–5426 (2022). https://doi.org/10.5194/essd-14-5411-2022

[9] M. Cranmer, "Interpretable machine learning for science with PySR and SymbolicRegression.jl," arXiv:2305.01582 (2023).

[10] S. L. Brunton, J. L. Proctor, and J. N. Kutz, "Discovering governing equations from data by sparse identification of nonlinear dynamical systems," *Proc. Natl. Acad. Sci. USA* **113**, 3932–3937 (2016). https://doi.org/10.1073/pnas.1517384113

TABLE IX. The eight eligible cells of the TEST length test. Measured: slope s (dB/km) and intercept I (dB) with 90 % intervals. Models: their own slope and intercept on the same values. Values inside the measured interval are marked with an asterisk.

Band	Class	Measured	No WAA	Const.	No mem.	Schleiss (b)	Pastorek (b)	Ours
Slope s (dB/km)								
28 GHz	0.5–1	0.037 [0.014, 0.077]	0.210	0.046*	0.098	0.042*	0.103	0.096
28 GHz	1–2	0.164 [0.086, 0.234]	0.374	0.179*	0.143*	0.188*	0.153*	0.168*
28 GHz	2–5	0.313 [0.205, 0.439]	0.708	0.453	0.273*	0.396*	0.266*	0.295*
28 GHz	5–10	0.603 [0.513, 0.863]	1.537	1.083	0.588*	0.984	0.552*	0.643*
38 GHz	0.5–1	0.150 [0.082, 0.221]	0.365	0.107*	0.201*	0.124*	0.200*	0.178*
38 GHz	1–2	0.332 [0.203, 0.455]	0.669	0.357*	0.322*	0.324*	0.333*	0.311*
38 GHz	2–5	0.598 [0.296, 0.787]	1.289	0.826	0.513*	0.689*	0.503*	0.522*
38 GHz	5–10	0.850 [0.121, 1.570]	2.839	2.045	1.129*	1.721	1.085*	1.187*
Intercept I (dB)								
28 GHz	0.5–1	1.35 [0.99, 1.52]	0.00	0.90	0.88	1.14*	0.87	0.80
28 GHz	1–2	1.43 [1.15, 1.71]	0.02	0.90	1.20*	1.11	1.22*	1.05
28 GHz	2–5	1.79 [1.22, 2.13]	0.07	0.90	1.82*	1.38*	1.71*	1.50*
28 GHz	5–10	2.60 [1.11, 3.47]	0.13	0.99	2.44*	1.46*	2.47*	2.11*
38 GHz	0.5–1	1.44 [1.11, 1.60]	0.01	0.93	0.93	1.18*	0.93	0.87
38 GHz	1–2	1.55 [1.20, 1.79]	−0.01	0.91	1.21*	1.23*	1.24*	1.09
38 GHz	2–5	2.00 [1.54, 2.31]	0.04	0.95	2.00*	1.57*	1.86*	1.66*
38 GHz	5–10	3.42 [1.01, 4.90]	−0.01	0.89	2.64*	1.68*	2.60*	2.28*